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Impact of Artificial Intelligence in Promoting Academic Integrity in Education: A Systematic review

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Abstract

Technological advancements are improving very fast, just recently there was the introduction of ChatGPT has had a serious impact on education. This poses a significant challenge to education integrity, accessibility, and inclusivity. Many students worldwide do online learning due to the pandemic that hit the world back in 2019. This led to integrating online as a strategy to control in-person interactions. Online learning has thus led to increased cases of cheating and a lack of academic integrity. Using Meta-Analysis, this study intends to determine how using artificial intelligence has helped promote academic integrity. This brief scoping study aims to gauge the depth of the literature exploring the relationship between academic honesty and AI in schools of higher learning. This review will examine the papers included to draw conclusions about this developing field and the ethical considerations that must be *considered. The higher education administration, faculty, administration, and those studying academic integrity will all find our findings helpful.*

Keywords: Higher Education, Academic Integrity, Artificial Intelligence, Meta-Analysis.

Introduction

Future advancements in higher education are inextricably intertwined with the increasing computing capacities of emerging intelligent machines. Technological progress in AI has created exciting new opportunities and challenges for higher education educators and has the potential to radically alter the structure and administration of today's universities (Mamani et al., 2022). There is little consensus on an ultimate definition of artificial intelligence because philosophical perspectives have impacted the answers since Aristotle (Andrea et al., 2015).

Since its inception in 1956, the concept of artificial intelligence (AI) has been the subject of a wide variety of theoretical interpretations from various disciplines, including chemistry, biology, linguistics, mathematics, and the advancement of AI technology. Despite the wealth of definitions and points of view, however, there is substantial disagreement (Andrews et al., 2016). Most approaches focus on the cognitive aspects of intelligence, ignoring or giving less weight to the political, psychological, and philosophical facets of intelligence. We give a working definition based on what we've learned through a literature review to aid in our analysis of the effects of AI on higher education. AI can be defined as computer systems that mimic human intelligence by performing activities including learning, adapting, synthesis, self-correction, and data utilization in complicated processing tasks like humans (Ramos, et al., 2022)

According to Bostrom (2016), using computer-generated writing in the academic setting raises several moral dilemmas and concerns. On the one hand, the employment of AI can compromise the effectiveness of evaluations of student learning and devalue a diploma. As shown by Bayne (2015), students may not learn the material well if they are not involved in the learning process and instead depend on AI to do all of the work for them. They may move forward with the following assignments unprepared. In addition, pupils may be able to plagiarize or cheat through AI without being caught. This violates standards of honesty in the classroom.

Conversely, when utilized properly and following established rules, AI can be useful for instruction and evaluation. The difficulty is that most schools do not have such policies in place. ChatGPT is an artificial intelligence (AI) tool that can increase productivity and precision while giving students more options for showcasing their abilities (Chen et al., 2015). To avoid unethical situations, however, AI shouldn't be employed to give some students an unfair advantage. Artificial intelligence (AI) can be both a boon and a bane in the classroom.

On the one hand, these resources are commonly utilized to back up the academic community and their efforts to combat plagiarism to preserve academic integrity and help students improve their writing to attain their full potential (Gavilán, et al., 2022). However, when AI improves and gets "smarter," students will be able to take advantage of these powerful tools to have them produce high-quality essays and articles for them. Is this a problem that schools are prepared to solve? Thus, this research intends to determine the impact of artificial intelligence in promoting academic integrity in education (Mansilla, et al., 2022)

This study starts with the introduction which provides a general overview of artificial intelligence and how these AIs promote academic integrity. The second section of this research is the literature review which discusses previous research on artificial intelligence and its relationship with integrity in academics in various institutions. The following section in this research is the methodology section which according to this study, is the Preferred Reporting Item for Systematic Reviews and Meta-Analyses (PRISMA) methodology. Finally, this study summarizes by providing a discussion and a conclusion.

Literature Review

Artificial intelligence, according to experts in the field of education, is "computing systems that can engage in human-like activities such as learning, adapting, synthesizing, self-correcting, and usage of data for complex processing tasks" (De Lange, 2015). What AI will entail for schools, laboratories, and standardized tests is yet murky and complicated (Diss, 2015; Gallagher, 2015). One example of an ever-improving AI that can produce writing that sounds natural with or without human intervention is the Generative Pre-Trained Transformer (GPT-3) (Gibney, 2017; GFKI, 2015). The most popular software today, for instance, uses artificial intelligence to rewrite entire sentences (Grove, 2015). Accessibility and inclusion are also being enhanced by advances in artificial intelligence such as text summarization, real-time captioning, machine translation, and built-in libraries of idioms and phrases (Kim, 2018). Under the UDL framework, technologies are considered neutral mediums that allow for a wide range of interaction, representation, and expression (Lunny et al., 2021)

In artificial intelligence, there is a distinction between "weak" AI and "strong" AI, sometimes known as "narrow" and "general" AI (Martinez, 2021). The philosophical topic of whether or not machines will be able to reason at all, let alone develop consciousness, remains unanswered. A powerful or universal AI is quite unlikely to develop anytime soon. For this reason, we are working with GOFAI ("good old-fashioned AI," a phrase coined by the philosopher John Haugeland in 1985 as cited by Grove, 2015) in the academic setting, where agents and information systems are treated as if they possessed intelligence.

The educational landscape is becoming increasingly cloudy and convoluted as AI becomes more pervasive (Castillo-Acobo et al., 2022; Arias et al., 2022). Implementing AI technology in higher education might present challenges, and many educators may lack the knowledge and training to address them (Martinez, 2021; Mindzak, 2020; Muñoz et al., 2022). Many schools are implementing intelligent tutoring systems (ITS) to provide students with an education on par with what they would receive in a private setting.

Data from learner models, algorithms, and neural networks can inform decisions on a learning path and material selection, cognitive scaffolding and support, and student-centered dialogue. As one-on-one human coaching is impractical in large-scale distance education institutions that run modules with thousands of students, ITS provides immense promise. Innumerable research has demonstrated the importance of social interaction and teamwork in the educational process (for example, Luckin, 2017; Maderer, 2016; Neven, 2013). Yet, online collaboration needs to be facilitated and moderated (Neven, 2015; Luckin, 2017; Puma et al., 2022). Because of its ability to support adaptive group formation based on learner models, AIEd can help with collaborative learning by either facilitating online group interaction or summarizing discussions that a human tutor can use to guide students toward the course's goals and objectives.

Additionally, students are actively engaged and directed through genuine VR and game-based learning environments using intelligent virtual reality (IVR), which draws on ITS. For instance, in remote or online laboratories, virtual agents can play the roles of instructors, guides, and even students' peers (Perez et al., 2016). Learning the intricacies of AI is a necessary skill for today's college educators. Some writers have emphasized the importance of distinguishing between AI's supportive and cheating functions and the necessity of human understanding of the differences between the two Lazarus et al., 2008). Academics have also stressed the importance of human problem-solving, criticism, and questioning, notwithstanding AI developments (Hillier, Write & Damen, 2015). A scholarly conversation on AI in higher education is necessary to inform further initiatives, as proposed by Munn et al. (2018). More crucially, in the contemporary higher education setting, understanding how this technology may affect academic integrity is paramount.

Method

A systematic review aims to provide definitive answers to research questions using a predetermined set of search terms and a predetermined set of inclusion and exclusion criteria that can be replicated (Gough, Oliver & Thomas, 2017). The next step is to extract and code data from the included research so that we may draw conclusions, shed light on how these findings might be put into practice, and identify any inconsistencies in the data. This work organizes 70 articles about AI in education and 26 are included in the study.

Eligibility criteria

Scope reviews will be evaluated according to the Population, Concept, and Context paradigm (Kübler et al., 2015; Morrison & Mindzak, 2021; Mindzak, 2020) to determine suitability. So that this brief scoping review can adequately inform readers and reviewers, we have devised broad inclusion criteria (Munn et al., 2018; Perez, 2016). This brief scoping research focuses on those who work in higher education, such as professors, teaching assistants, tutors, and instructional designers. No distinction will be made between the various ranks of professors, including Professor, Associate Professor, Assistant Professor, and Lower-Ranked Professor. The large population comprises students enrolled in Canadian post-secondary institutions (Government of Canada, 2022). Teaching assistant positions are often given to students with relevant work experience (Education USA, n.d.). Academic student support staff in higher education serve as a bridge between instructors and students needing professional resources.

The final group, educational developers, consists of individuals who work with professors, academic programs, and other campus organizations to improve the quality of education offered (Reuters, 2016). All ages and sexes are welcome to join in. Uncertain studies that could have included any of these people will be disqualified. The primary condition is that these stakeholders have duties directly associated with teaching and learning in higher education (Lunny et al., 2021; Perez, 2016 Based on the advice of Munn et al. (2018), we hope to add to the body of literature by providing a professor's perspective on the use of AI in higher education.

Search Strategy

Using the original search phrase and parameters provided in Tables 1 and 2, we looked for English-language, peer-reviewed papers reporting AI in education in EBSCO Education Source, Web of Science, and Scopus (covering titles, abstracts, and keywords). Although there are valid concerns about the peer-review process in the scientific community (for example, Smith 2006), this assessment only considers works published in scholarly peer-reviewed journals (Nicholas et al., 2015). In the first month of the search, 2023, we came across 5,980 distinct records.

Table 1: Original Search criteria

Research Topics	Search Terms
Artificial intelligence in Education	"Higher education," "Artificial intelligence or machine learning," "Chatbots," "automated tutor," "student or personal tutor," "undergraduate or college," or "adult education"
Learning settings	Students or Learning process

Table 2: Inclusion Criteria

Criteria for Inclusion	Criteria for exclusion
English	The article is not written in English
Higher Education	Article not done on higher education
Published between 2019 to 2022	The publication did before 2019
Empirical or Primary methodology used	The methodology used is not primary or empirical
Published in Scopus, EBSCO, Education Source or Wed of Science	The study conducted is a journal article
Application of Artificial intelligence in Education	Not an artificial intelligence study/or an education or learning setting

Preferred Reporting Items for Systematic Reviews and Meta-Analyses PRISMA Methodology

Three coders reviewed the titles and abstracts of 5,980 items, prioritizing sensitivity over specificity (i.e., including rather than excluding articles). Discussions were held at length to settle on criteria for including or eliminating the initial 75 articles. Martinez's study demonstrated inter-rater reliability by comparing the coding decisions of three coders (A, B, and C) across 20 randomly selected articles (2021). Several codes in the coding scheme significantly predicted the coefficient representing inter-rater reliability (Nicholas, 2015). Kappa levels between 0.71 to 0.80 are considered fairly good, while values of 0.80 and above are exceptional (Lunny et al., 2021).

From the initial pool of articles, 205 were selected for full-text screening (see Fig. 1). Only 41 papers were retrievable through the library order mechanism or by contacting the authors directly. To synthesize the data, we retrieved, vetted, and coded 70 articles, eliminating 26 papers along the way.

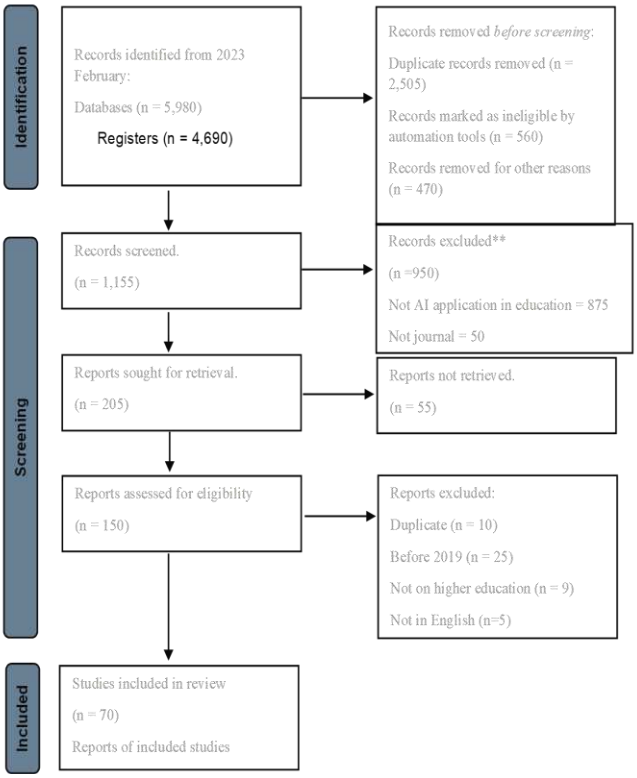

Figure 1: Prisma Diagram

Note: source is from: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement.

Table 3: Articles used

Reference	Year of publication	Technic used	Limitation of the study
Arias Gonzales et al.	2022	Quantitative	The study's findings may not be generalizable to the population as a whole because of the small size of the sample used. A potentially limiting factor is the study's likely narrow focus on one subset of public sector personnel. The reliability of the results may have been compromised by the study's possible reliance on self-reported data, which is susceptible to social desirability bias.

Castillo-Acobo et al.	2022	Qualitative	The study depends on instructors' self-reported data, which could be skewed by social desirability bias or memory recall bias. The study was only conducted in Venezuela, thus the results may not generalize to other countries with vastly different historical, political, social, and educational backgrounds.
Lunny et al.	2021	Meta-Analysis	As the study just aimed to summarize the evidence, rather than do a meta-analysis, it may have limited data synthesis. It's possible, then, that the results don't give us a full picture of the reliability and efficiency of dose-sparing methods.
Mamani et al.	2022	Quantitative	The study used a cross-sectional design, which makes it difficult to deduce causes and effects. The study does not establish a cause-and-effect relationship between sustainable development and entrepreneurialism or green innovation aim.
Martínez	2021	Not stated	The article focuses only on a few examples of AI applications for people with disabilities, such as voice assistants and image recognition software. The author does not provide a comprehensive overview of the field, and there may be other areas where AI could be applied to improve accessibility.
Mindzak	2020	Not stated	The article does not present any empirical research or data to support the claims made about the potential impact of AI on academic writing. The author only provides opinions and anecdotes to illustrate the potential implications of AI-generated writing.
Morrison & Mindzak	2021	Meta-Analysis	Due to time limits or a lack of extensive research on the topic, the webinar may not have covered all the potential effects of text-generating technology on academic integrity. Furthermore, the webinar's ideas and views might not be representative of the wide range of people and organizations whose lives have been altered by new innovations. Yet, there may have been restrictions on audience participation that prevented the webinar from eliciting additional in-depth thoughts and viewpoints on the topic. As a final point, given that the webinar was sponsored by a single university, it's possible that the ideas and arguments offered there won't apply to other schools or situations.

Muñoz et al.	2022	Meta-Analysis	This investigation may have been strengthened by including data from lower-level educational institutions as well as universities. Because of this, the results cannot be applied to any other school. In addition, there may have been studies that were not included because they did not meet the inclusion and exclusion criteria, which would have strengthened the study's ability to provide a more all-encompassing analysis of the topic.
Page et al.	2020	Meta-Analysis	The quality of reporting in systematic reviews may suffer if authors and publications opt to disregard or just partially comply to the guideline. The PRISMA 2020 statement should be used in conjunction with other applicable guidelines and standards because it may not address all potential concerns or challenges that can occur during the conduct and reporting of a systematic review.
Puma et al.	2022	Qualitative	location biasness

Result

Using the concept of the student life cycle as a framework, several AI-based services were defined at the institutional and administrative level (such as admission, counseling, and library services) and the academic support level for teaching and learning (e.g., assessment, feedback, tutoring). As can be seen in the table below, 42 (60.0%) of the studies dealt with academic support services, 20 (29.0%) dealt with administrative and institutional services, and 8 (11.0%) dealt with both.

Table 4: Case studies for the selected articles

Case study for selected articles	Distribution	
1. Academic support	42	60%
2. Administration and Institutional services	20	29%
both 1 and 2	8	11%

Fifty studies (71.3% of the total) were designed for undergraduates, eleven (26.0%) were aimed at graduate students, and five (2.7%) did not specify the degree of study as shown in the figure below.

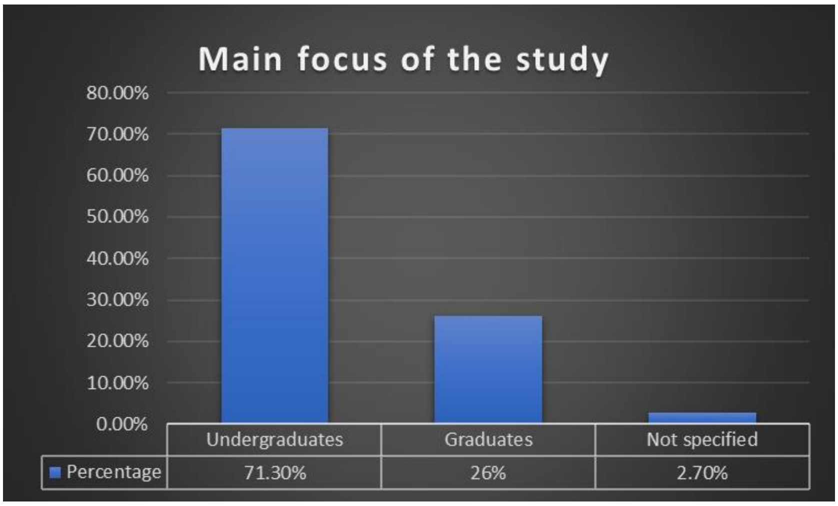

Figure 2: Main focus of the study for the articles selected

Table 5: Methodology used in the selected articles

Methodology used	**Distribution**
Theoretical/descriptive	18
Quantitative research technic	51
qualitative	1
Hybrid strategy	1
Total	70

Twenty-five percent, or 18 studies, were considered theoretical or descriptive. Seventy-three percent (51 articles) relied solely on quantitative techniques, while 1 percent employed qualitative and 1 percent employed a hybrid strategy. This qualitative study aimed to compare the feedback that ESL students received from a computerized essay grader and a human teacher (Gibney, 2017). For their studies, many authors turned to quasi-experimental designs, which involve randomly assigning participants to either an experimental group in which an AI application (like an intelligent tutoring system) was applied or a control group in which no intervention was applied and then testing both groups before and after.

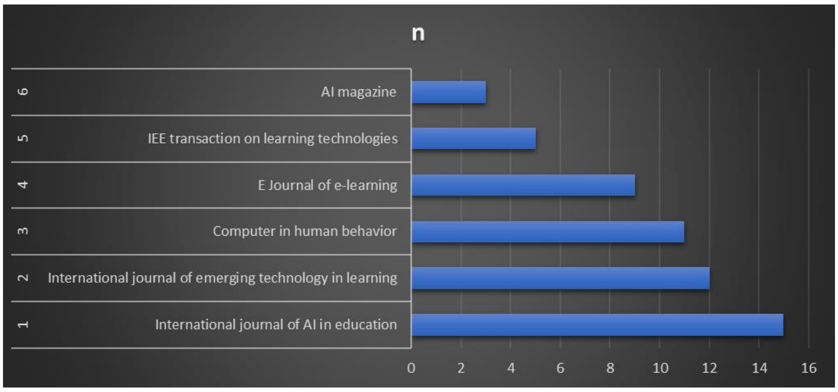

Figure 3: Publication trend

As shown in the figure above, there is an increasing trends in AI research in various publication sites, this makes it a key factor or cause of concern for many institutions like the government and education institutions as the future of education has been changing with the changing rate of technological advancement every year.

Conclusion

The rapid rise of AI requires universities and colleges to reevaluate their approach to education, including the roles of instructors and pedagogical strategies. There are concerns about the loss of jobs, individual liberty, and the emergence of a dystopian state due to the growth of tech lords and digital firms. Higher education institutions should include these risks, along with others like climate change and natural resource depletion, in their long-term planning. The results of a scoping review can be used by faculty members to discuss the implications of AI-generated writing for academic integrity with administrators, peers, and students. Collaboration between professors, TAs, academic aides, ADA experts, course designers, and librarians can ensure that students have access to innovative courses that teach them how to use AI writing tools in a fair and just manner. As universities adopt AI in the classroom, they need to reevaluate their roles and future relationships with companies that provide AI solutions and their owners to protect the higher education system's mission.

In other words, colleges and universities need to reevaluate their roles, pedagogical approaches, and future relationships with companies that provide artificial intelligence (AI) solutions and their owners. And universities foresee a long list of opportunities and difficulties ahead as they adopt AI in the classroom. These options provide fresh possibilities for teaching and learning in a way that benefits everyone while also laying the groundwork for a more robust model that can protect fundamental principles and the higher education system's mission.

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